

Nucleus Specifications

Hardware specifications, Nucleus mesh radio.



Compute

Item	Specification
Platform	Raspberry Pi 4
Operating system	Raspberry Pi OS

Radios

Item	Specification
WiFi mesh radio	2.4 GHz, 20 MHz channel
WiFi mesh conducted output power	500 mW
LoRa mesh radio	915 MHz, SF7, 250 kHz bandwidth
LoRa mesh conducted output power	1,000 mW

I/O

Item	Specification
WiFi access point	5 GHz
Ethernet	10/100/1000 Mbps, RJ45
USB-C	USB 2.0

Antennas

Item	Specification
WiFi mesh antenna	5 dBi
LoRa mesh antenna	5 dBi
Connector type	RP-SMA

Power

Item	Specification
Input voltage	5 V DC
Average current draw	0.6 A
Typical power consumption	3 W
Battery	External

Enclosure

Item	Specification
Material	PCTG

Range

Range on a mesh network is governed by node density more than by the output power of any single unit. Coverage is extended by adding nodes, not by increasing power at one of them. All figures below are single links between two radios, and are drawn from the field tests documented on the [Test Data](#) page.

WiFi mesh	Measured range
50 m elevation difference, clear line of sight	5 km
Ground level	1 km
Heavy obstruction, buildings or dense tree cover	100 m

LoRa	Measured range
With elevation, longest test conducted	22 miles, link held
With elevation, secondary test	8 miles, link held
Maximum range	Not established in testing to date

LoRa results were measured on 159 mW radios. V3 ships a 1,000 mW LoRa radio, not yet range tested.